



ROTARY DOUBLE INVERTED PENDULUM MODELING & CONTROL

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Undergraduate Research

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Abstract

This report presents the mathematical modeling and control design of a rotary double inverted pendulum. A nonlinear model is derived using Lagrangian mechanics and linearized about the upright equilibrium. State-space methods are then used to design stabilizing controllers, and the performance of pole-placement and LQR controllers is evaluated through simulation and experimental validation.

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1 Nomenclature

θ Rotary arm angular position [rad]

α First pendulum angular position [rad]

ϕ Second pendulum angular position [rad]

$\dot{\theta}$ Rotary arm angular velocity [rad/s]

$\dot{\alpha}$ First pendulum angular velocity [rad/s]

$\dot{\phi}$ Second pendulum angular velocity [rad/s]

$\ddot{\theta}$ Rotary arm angular acceleration [rad/s²]

$\ddot{\alpha}$ First pendulum angular acceleration [rad/s²]

$\ddot{\phi}$ Second pendulum angular acceleration [rad/s²]

V_m Input motor voltage [V]

r Rotary arm length [m]

l_1 Distance from the first pendulum joint to its centre of mass [m]

l_2 Distance from the second pendulum joint to its centre of mass [m]

L_r Total length of the rotary arm [m]

L_1 Total length of the first pendulum [m]

L_2 Total length of the second pendulum [m]

m_r Mass of the rotary arm [kg]

m_1 Mass of the first pendulum [kg]

m_h Mass of the hinge [kg]

m_2 Mass of the second pendulum [kg]

J_r Mass moment of inertia of the rotary arm about the fixed frame [kg·m²]

J_1 Mass moment of inertia of the first pendulum about its centre of mass [kg·m²]

J_2 Mass moment of inertia of the second pendulum about its centre of mass [kg·m²]

g Gravitational acceleration [m/s²]

τ Applied torque at the rotary arm [N·m]

2 DBIP Diagram - Quanser

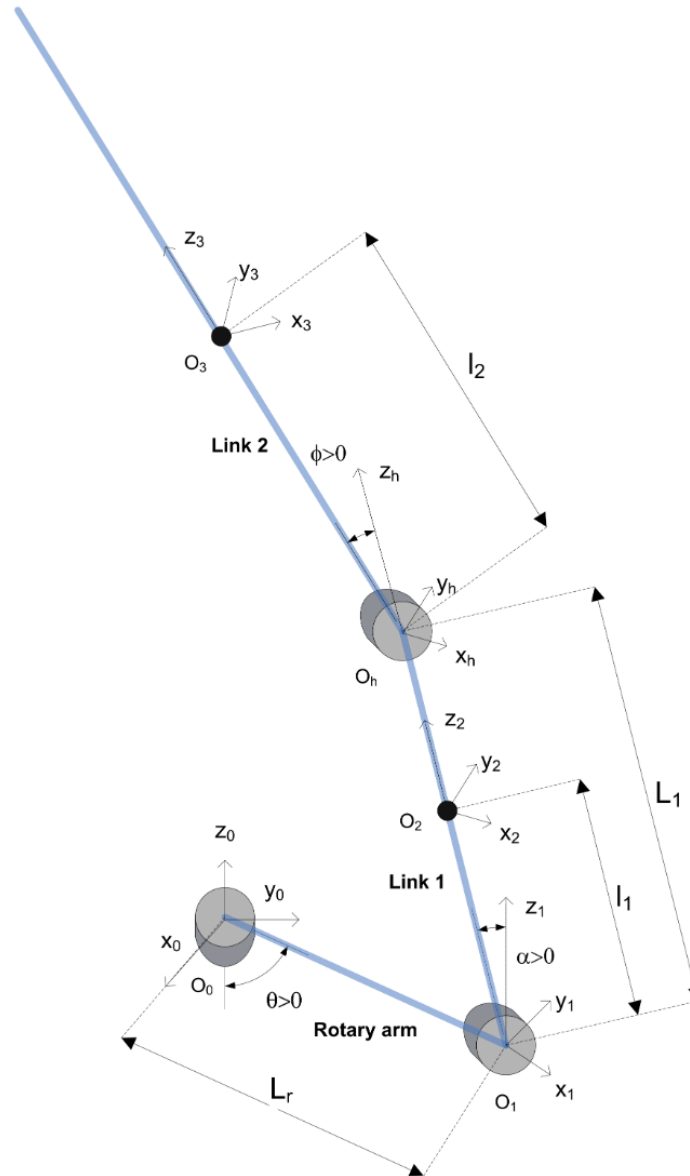


Figure 1: DBIP

3 Problem Statement & Definitions

In the following modeling of the system, we assume the mass are concentrated about the center of mass of the respective geometries and that they are rigid bodies. The frames of the DBIP are defined as follows:

- Frame 0 : Inertial Frame
- Frame 1 : Tip of the Arm
- Frame 2: Tip of the Bottom Link (Hinge)
- Frame 3: Tip of the Top Link

3.1 Rotation Matrices

Frame 1 relative to Frame 0:

$$\mathbf{R}_1^0 = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Frame 2 relative to Frame 1:

$$\mathbf{R}_2^1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\alpha) & -\sin(\alpha) \\ 0 & \sin(\alpha) & \cos(\alpha) \end{bmatrix}$$

Frame 3 relative to Frame 2:

$$\mathbf{R}_3^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi) \end{bmatrix}$$

3.2 Position Vectors

This subsection gives the resultant position vectors of the tip of the arm, center of mass of link 1, the hinge, the center of mass of link 2 relative to the inertial frame.

$$\mathbf{r}_{arm}^0 = \begin{bmatrix} L_r \cos(\theta) \\ L_r \sin(\theta) \\ 0 \end{bmatrix}$$

$$\mathbf{r}_1^0 = \begin{bmatrix} L_r \cos(\theta) + l_1 \sin(\alpha) \sin(\theta) \\ L_r \sin(\theta) - l_1 \sin(\alpha) \cos(\theta) \\ l_1 \cos(\alpha) \end{bmatrix}$$

$$\mathbf{r}_{hinge}^0 = \begin{bmatrix} L_r \cos(\theta) + L_1 \sin(\alpha) \sin(\theta) \\ L_r \sin(\theta) - L_1 \sin(\alpha) \cos(\theta) \\ L_1 \cos(\alpha) \end{bmatrix}$$

$$\mathbf{r}_2^0 = \begin{bmatrix} \sin(\theta) (L_1 \sin(\alpha) + l_2 \cos(\alpha) \sin(\phi) + l_2 \cos(\phi) \sin(\alpha)) + L_r \cos(\theta) \\ L_r \sin(\theta) - \cos(\theta) (L_1 \sin(\alpha) + l_2 \cos(\alpha) \sin(\phi) + l_2 \cos(\phi) \sin(\alpha)) \\ L_1 \cos(\alpha) + l_2 \cos(\alpha) \cos(\phi) - l_2 \sin(\alpha) \sin(\phi) \end{bmatrix}$$

3.3 Velocity Vectors

The velocity vectors are obtained by applying the Jacobian (first-order Taylor expansion of a multivariable function)

$$\dot{\mathbf{r}} = \frac{\partial \mathbf{r}}{\partial \mathbf{q}} \dot{\mathbf{q}}$$

where

$$\mathbf{r} = \begin{bmatrix} r_x \\ r_y \\ r_z \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} \theta \\ \alpha \\ \phi \end{bmatrix}$$

$$\dot{\mathbf{r}} = \begin{bmatrix} \dot{r}_x \\ \dot{r}_y \\ \dot{r}_z \end{bmatrix} = \begin{bmatrix} \frac{\partial r_x}{\partial \theta} & \frac{\partial r_x}{\partial \alpha} & \frac{\partial r_x}{\partial \phi} \\ \frac{\partial r_y}{\partial \theta} & \frac{\partial r_y}{\partial \alpha} & \frac{\partial r_y}{\partial \phi} \\ \frac{\partial r_z}{\partial \theta} & \frac{\partial r_z}{\partial \alpha} & \frac{\partial r_z}{\partial \phi} \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{\alpha} \\ \dot{\phi} \end{bmatrix}$$

The derived velocity vectors are:

$$\dot{\mathbf{r}}_{arm}^0 = \begin{bmatrix} -L_r \sin(\theta) & 0 & 0 \\ L_r \cos(\theta) & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{\alpha} \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} -L_r \dot{\theta} \sin(\theta) \\ L_r \dot{\theta} \cos(\theta) \\ 0 \end{bmatrix}$$

$$\dot{\mathbf{r}}_1^0 = \begin{bmatrix} l_1 \sin(\alpha) \cos(\theta) - L_r \sin(\theta) & l_1 \cos(\alpha) \sin(\theta) & 0 \\ L_r \cos(\theta) + l_1 \sin(\alpha) \sin(\theta) & -l_1 \cos(\alpha) \cos(\theta) & 0 \\ 0 & -l_1 \sin(\alpha) & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{\alpha} \\ \dot{\phi} \end{bmatrix}$$

$$= \begin{bmatrix} \dot{\alpha} l_1 \cos(\alpha) \sin(\theta) - \dot{\theta} (L_r \sin(\theta) - l_1 \sin(\alpha) \cos(\theta)) \\ \dot{\theta} (L_r \cos(\theta) + l_1 \sin(\alpha) \sin(\theta)) - \dot{\alpha} l_1 \cos(\alpha) \cos(\theta) \\ -\dot{\alpha} l_1 \sin(\alpha) \end{bmatrix}$$

$$\begin{aligned}
\dot{\mathbf{r}}_h^0 &= \begin{bmatrix} L_1 \sin(\alpha) \cos(\theta) - L_r \sin(\theta) & L_1 \cos(\alpha) \sin(\theta) & 0 \\ L_r \cos(\theta) + L_1 \sin(\alpha) \sin(\theta) & -L_1 \cos(\alpha) \cos(\theta) & 0 \\ 0 & -L_1 \sin(\alpha) & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{\alpha} \\ \dot{\phi} \end{bmatrix} \\
&= \begin{bmatrix} L_1 \dot{\alpha} \cos(\alpha) \sin(\theta) - \dot{\theta} (L_r \sin(\theta) - L_1 \sin(\alpha) \cos(\theta)) \\ \dot{\theta} (L_r \cos(\theta) + L_1 \sin(\alpha) \sin(\theta)) - L_1 \dot{\alpha} \cos(\alpha) \cos(\theta) \\ -L_1 \dot{\alpha} \sin(\alpha) \end{bmatrix} \\
\dot{\mathbf{r}}_2^0 &= \mathbf{J}_2(\mathbf{q}) \dot{\mathbf{q}} = \begin{bmatrix} \dot{r}_x \\ \dot{r}_y \\ \dot{r}_z \end{bmatrix},
\end{aligned}$$

4 Potential Energy of the System

The potential energies of the bottom link, hinge and the top link are respectively defined as:

$$V_1 = m_1 g r_{1z}^0$$

$$V_h = m_h g r_{hz}^0$$

$$V_2 = m_2 g r_{2z}^0$$

$$V_t = V_1 + V_h + V_2$$

4.1 Total Potential Energy

$$V_t = g m_2 (l_2 \cos(\alpha + \phi) + L_1 \cos(\alpha)) + L_1 g m_h \cos(\alpha) + g l_1 m_1 \cos(\alpha)$$

5 Kinetic Energy of the System

This section defines the kinetic energy of the rotary arm, bottom link, hinge and top link. The kinetic energy of a rigid object is the sum of two terms, the translational kinetic energy obtained by concentrating the entire mass of the object at the center of mass, and the rotational kinetic energy of the body about the center of mass

$$KE_{\text{total}} = KE_{\text{trans}} + KE_{\text{rot}} = \frac{1}{2} m \mathbf{v}^T \mathbf{v} + \frac{1}{2} \boldsymbol{\omega}^T \mathcal{J} \boldsymbol{\omega}$$

Where m is the total mass of the rigid body, v and ω are the linear and angular velocity vectors relative to the inertial frame.

$\mathcal{J}$ is the inertia tensor defined as

$$\mathcal{J} = RIR^T \quad (1)$$

Where I is the inertia tensor expressed in the body-attached frame and R is the orientation transformation from the body-attached frame and the inertial frame.

5.1 Addition of Angular Velocities

The addition of angular velocities will become important towards computing the rotational kinetic energy of the bottom and top link. That is, we investigate the resultant angular velocity due to the relative rotation of several coordinate frames[2]. In the general case, a time-varying rotation matrix R is defined as

$$\dot{R} = S(\omega(t))R(t) \quad (2)$$

Where $S(\omega(t))$ is skew symmetric and $R(t)$ is the angular velocity of the rotating frame relative to the fixed frame at time t . In the following, the notation $\omega_{i,j}$ denote the angular velocity vector that corresponds to the time derivative of the rotation matrix R_j^i . The derivations from the book [2] leads to the formula:

$$\omega_{0,n}^0 = \omega_{0,1}^0 + R_1^0 \omega_{1,2}^1 + R_2^0 \omega_{2,3}^2 + R_3^0 \omega_{3,4}^3 + \dots + R_{n-1}^0 \omega_{n-1,n}^{n-1} \quad (3)$$

We are interested in $\omega_{0,2}^0$ and $\omega_{0,3}^0$ which are the angular velocities for the frames aligned with the bottom and top link respectively relative to the inertial frame.

5.2 Kinetic Energy of the Arm

$$\begin{aligned} KE_{total} &= \frac{1}{2}m(\omega \cdot d)^2 + \frac{1}{2}I \cdot \omega^2 \\ &= \frac{1}{2}\omega^2(I + md^2) \end{aligned} \quad (4)$$

Where the right term is the parallel axis theorem for the inertia¹ about the axis of rotation relative to the inertial frame. q_1 refers to $\omega = \dot{\theta}$.

5.3 Translational Kinetic Energy of Link 1

Take the derived velocity vector and compute:

$$T_{t1} = \frac{1}{2}m_1\dot{r}_{01}^2$$

¹The value given by Quanser's documentation is erroneous, look at the values I give later in this document

5.4 Rotational Kinetic Energy of Link 1

The inertia tensor about the body attached frame is shown as follows. Note that we have no I_{zz} entry because we assume it's a thin rod.

$$I_{link1} = \begin{bmatrix} \frac{1}{12}m_1L_1^2 & 0 & 0 \\ 0 & \frac{1}{12}m_1L_1^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The inertia tensor $\mathcal{J}$ obtained through the similarity transformation as seen in (1):

$$\mathcal{J}_{link1} = R_2^0 \cdot I_{link1} \cdot R_2^{0T}$$

$$\begin{bmatrix} \cos(\theta) & -\cos(\alpha)\sin(\theta) & \sin(\alpha)\sin(\theta) \\ \sin(\theta) & \cos(\alpha)\cos(\theta) & -\sin(\alpha)\cos(\theta) \\ 0 & \sin(\alpha) & \cos(\alpha) \end{bmatrix} \begin{bmatrix} \frac{1}{12}m_1L_1^2 & 0 & 0 \\ 0 & \frac{1}{12}m_1L_1^2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \cos(\theta) & -\cos(\alpha)\sin(\theta) & \sin(\alpha)\sin(\theta) \\ \sin(\theta) & \cos(\alpha)\cos(\theta) & -\sin(\alpha)\cos(\theta) \\ 0 & \sin(\alpha) & \cos(\alpha) \end{bmatrix}^T$$

The angular velocity of the rotating frame 2 relative to the inertial frame 0 using the formula from (3):

$$\begin{aligned} \omega_{0,2}^0 &= \omega_{0,1}^0 + R_1^0 \omega_{1,2}^1 \\ &= \begin{bmatrix} 0 \\ 0 \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\alpha} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \dot{\alpha} \cos(\theta) \\ \dot{\alpha} \sin(\theta) \\ \dot{\theta} \end{bmatrix} \end{aligned}$$

The rotational kinetic energy of the bottom link is finally given as:

$$\begin{aligned} T_{r1} &= \frac{1}{2} \omega_{0,2}^{0T} \cdot \mathcal{J}_{link1} \cdot \omega_{0,2}^0 \\ &= \frac{L_1^2 m_1}{48} (2\dot{\alpha}^2 - \dot{\theta}^2 \cos(2\alpha) + \dot{\theta}^2) \end{aligned}$$

5.5 Kinetic Energy of the Hinge

The translational kinetic energy of the hinge is described as:

$$T_{htrans} = \frac{1}{2} m_h \dot{r}_{0h}^2$$

The latter doesn't have rotational kinetic energy because it is treated as a point mass.

5.6 Translational Kinetic Energy of Link 2

Take the derived velocity vector and compute:

$$T_{t2} = \frac{1}{2} m_2 \dot{r}_{02}^2$$

5.7 Rotational Kinetic Energy of Link 2

The inertia tensor about the body attached frame is shown as follows. Note that we have no I_{zz} entry because we assume it's a thin rod. The inertia tensor is:

$$I_{link2} = \begin{bmatrix} \frac{1}{12} m_2 L_1^2 & 0 & 0 \\ 0 & \frac{1}{12} m_2 L_1^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The inertia tensor $\mathcal{J}$ obtained through the similarity transformation as seen in (1):

$$\begin{aligned} \mathcal{J}_{link2} &= R_3^0 \cdot I_{link2} \cdot R_3^{0T} \\ &= \begin{bmatrix} \cos(\theta) & -\cos(\alpha + \phi) \sin(\theta) & \sin(\alpha + \phi) \sin(\theta) \\ \sin(\theta) & \cos(\alpha + \phi) \cos(\theta) & -\sin(\alpha + \phi) \cos(\theta) \\ 0 & \sin(\alpha + \phi) & \cos(\alpha + \phi) \end{bmatrix} \begin{bmatrix} \frac{1}{12} m_2 L_2^2 & 0 & 0 \\ 0 & \frac{1}{12} m_2 L_2^2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \cos(\theta) & -\cos(\alpha + \phi) \sin(\theta) & \sin(\alpha + \phi) \sin(\theta) \\ \sin(\theta) & \cos(\alpha + \phi) \cos(\theta) & -\sin(\alpha + \phi) \cos(\theta) \\ 0 & \sin(\alpha + \phi) & \cos(\alpha + \phi) \end{bmatrix}^T \end{aligned}$$

The angular velocity of the rotating frame 3 relative to the inertial frame 0 using the fomrula from (3):

$$\omega_{0,3}^0 = \omega_{0,1}^0 + R_1^0 \omega_{1,2}^1 + R_2^0 \omega_{2,3}^2$$

$$= \begin{bmatrix} 0 \\ 0 \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\alpha} \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \cos(\theta) & -\cos(\alpha) \sin(\theta) & \sin(\alpha) \sin(\theta) \\ \sin(\theta) & \cos(\alpha) \cos(\theta) & -\sin(\alpha) \cos(\theta) \\ 0 & \sin(\alpha) & \cos(\alpha) \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos(\theta)(\dot{\alpha} + \dot{\phi}) \\ \sin(\theta)(\dot{\alpha} + \dot{\phi}) \\ \dot{\theta} \end{bmatrix}$$

The rotational kinetic energy of the top link is finally given as:

$$\begin{aligned} T_{r2} &= \frac{1}{2} \cdot \omega_{0,3}^{0T} \cdot \mathcal{J}_{link2} \cdot \omega_{0,3}^0 \\ &= \frac{L_2^2 m_2}{48} (4\dot{\alpha}\dot{\phi} - \dot{\theta}^2 \cos(2\alpha + 2\phi) + 2\dot{\alpha}^2 + 2\dot{\phi}^2 + \dot{\theta}^2) \end{aligned}$$

5.8 Sum of Kinetic Energy

$$T_t = T_{arm} + T_{1trans} + T_{r1} + T_{htrans} + T_{2trans} + T_{r2}$$

6 Lagrangian Formulation

Using MATLAB symbolic toolkit, we simplify the Lagrange equation

$$L(q, \dot{q}) = T_t - V_t \quad (5)$$

6.1 Generalized Forces

The generalized forces include the torque and the damping coefficients:

$$\tau = \begin{bmatrix} \tau_1 - D_r \dot{\theta} \\ -D_1 \dot{\alpha} \\ -D_2 \dot{\phi} \end{bmatrix}$$

where

$$\tau = \frac{\eta_g K_g \eta_m k_t (V_m - K_g k_m \dot{\theta})}{R_m} 2$$

6.2 Euler Lagrange Equation

The Euler Lagrange Equation is of the form:

$$\frac{\partial^2 L}{\partial t \partial \dot{q}} - \frac{\partial L}{\partial q} = Q \quad (6)$$

We solve symbolically on MATLAB:

$$\frac{\partial^2 L}{\partial t \partial \dot{q}} - \frac{\partial L}{\partial q} - Q_i = 0$$

And solve for $\ddot{\theta}, \ddot{\alpha}, \ddot{\phi}$. Thus we now have a nonlinear state space model of the form:

$$\dot{x} = f(x, u) = [\dot{\theta}, \dot{\alpha}, \dot{\phi}, \ddot{\theta}, \ddot{\alpha}, \ddot{\phi}]^T \quad (7)$$

With

$$x = [\theta, \alpha, \phi, \dot{\theta}, \dot{\alpha}, \dot{\phi}]^T \quad (8)$$

6.3 Linearization - ABCD Realization

The controlled equilibrium point is the zero x vector with no voltage input. We compute the Jacobians.

$$x_{eq} = [\theta \ 0 \ 0 \ 0 \ 0 \ 0]^T \quad \text{and} \quad v_{eq} = 0$$

Take notice that the documentation provided by Quanser has many mistakes such as providing the wrong inertia value for the arm and discrepancies in how they apply formulas to derive the matrix realizations.

²Take into account take Quanser assume η_g & η_m are equal to 1

The errors in Quanser's documentation are as follows:

- J_r 's inertia value
- Mixup between top and bottom link parameters in the MATLAB script
- Don't account for efficiency parameters η_g, η_m
 - I just set those values to 1, it works but we could experiment using the actual efficiency values to see if they make a difference
- Double count of the R_m for the back-emf contribution to the A matrix calculation in their MATLAB script
- On their Maple derivation documentation, they neglect damping friction but on the MATLAB script they do account for it

The updated table of values is as follows:

Table 1: DBIP Updated Parameter Values

Parameter	Experimental (Correct)	Quanser's
L_r	0.2159	0.2159
L_1	0.2000	0.2000
l_1	0.1635	0.1635
L_2	0.3365	0.3365
l_2	0.1778	0.1778
m_1	0.097	0.097
m_h	0.141	0.1410
m_2	0.127	0.1270
J_r	9.9829E-04	0.0041
η_g	1	0.90
η_m	1	0.69
K_g	70	70
k_t	7.68E-3	7.68E-3
k_m	7.68E-3	7.68E-3
R_m	2.6	2.6

Computing the AB matrices using the jacobian and evaluating at the point $(x^*, u^*) = (x_{eq}, v_{eq})$ with:

$$A = \frac{\partial f}{\partial x}|_{(x^*, u^*)} \quad B = \frac{\partial f}{\partial u}|_{(x^*, u^*)}$$

$$A = \begin{bmatrix} 0 & 0 & 0 & 1.0000 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1.0000 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1.0000 \\ 0 & 493.7629 & -3.5865 & -74.8161 & -1.7518 & 1.7906 \\ 0 & 597.3631 & -23.7073 & -82.8864 & -2.1875 & 2.4444 \\ 0 & -610.6139 & 83.3820 & 84.7249 & 2.4444 & -3.3478 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 136.2256 \\ 150.9200 \\ -154.2677 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \quad D = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

7 Controllability and Pole Placement

This sections deals with the stabilization of the DBIP around the upright linearized point. If a system is controllable, the input has enough authority to steer the plant at any state at some point in time. Thus we begin by testing for controllability by checking the rank of the following matrix:

$$\text{rank} \begin{bmatrix} B & AB & A^2B & \dots & A^{n-1}B \end{bmatrix} \quad (9)$$

An example of a working set of desired eigenvalues is:

$$\sigma(A) = \{-109, -9 + 1.62j, -9 + 1.62j, -5, -4, -0.2\}$$

Using the place() function on MATLAB, we obtain the gain vector:

$$F = [-0.08989 \quad 11.082 \quad 40.481 \quad 0.037527 \quad 4.8218 \quad 5.1123]$$

8 Observability

This section addresses **state estimation**, since the physical system only provides direct measurements of the joint positions through the encoders. The corresponding joint velocities are not measured directly and are instead obtained by differentiating the position measurements and filtering the resulting signals using a second-order-low-pass filter. This motivates the use of observers and follow the procedure from [1, p. 125].

8.1 Full Order Observer Design - DBIP System

We are tasked with designing a full order observer for the system. First check if the pair $(\mathbf{C}, \mathbf{A})$ is observable which in our case is. An example of a full order observer has the form

$$\begin{aligned}\dot{z}(t) &= (A + LC)z(t) + (B + LD)u(t) - Ly(t) \\ \hat{x}(t) &= z(t)\end{aligned}$$

Where we pick L such that $A + LC$ is stable. For our purposes we pick the poles $\sigma = \{-121, -122, -123, -124, -125, -126\}$ The inputs to the dynamic observer will be

$$\begin{bmatrix} u(t) \\ y(t) \end{bmatrix}$$

And will have the state space form

$$\left[\begin{array}{c|cc} A + LC & B + LD & -L \\ \hline I & 0 & 0 \end{array} \right]$$

Giving

$$A_o = \begin{bmatrix} -172.32 & 1.5809 & -1.7248 & 1 & 0 & 0 \\ 87.911 & -243.25 & -3.973 & 0 & 1 & 0 \\ -83.815 & -3.7798 & -245.08 & 0 & 0 & 1 \\ -2677.6 & 280.35 & -294.98 & -74.816 & -1.7518 & 1.7906 \\ 14507 & -14666 & -636.14 & -82.886 & -2.1875 & 2.4444 \\ -13990 & -612.45 & -14763 & 84.725 & 2.4444 & -3.3478 \end{bmatrix}$$

$$B_o = \begin{bmatrix} 0 & 172.32 & -1.5809 & 1.7248 \\ 0 & -87.911 & 243.25 & 3.973 \\ 0 & 83.815 & 3.7798 & 245.08 \\ 136.23 & 2677.6 & 213.41 & 291.39 \\ 150.92 & -14507 & 15263 & 612.43 \\ -154.27 & 13990 & 1.8385 & 14847 \end{bmatrix}$$

$$C_o = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$D_o = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

8.2 Minimum Order Observer Design - DBIP System

We are tasked with designing a minimal order observer for the system

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 493.7629 & -3.5865 & -74.8161 & -1.7518 & 1.7906 \\ 0 & 597.3631 & -23.7073 & -82.8864 & -2.1875 & 2.4444 \\ 0 & -610.6139 & 83.3820 & 84.7249 & 2.4444 & -3.3478 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 136.2256 \\ 150.9200 \\ -154.2677 \end{bmatrix}$$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} x.$$

We can check for observability through the matrix³:

$$\mathcal{O}(\text{observable}) \Leftrightarrow \text{rank} \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^5 \end{bmatrix} = 6.$$

The pair (C, A) is observable and following the procedure we have:

Step 1. Finding T^{-1}

$$\text{Ker } C = \text{span} \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Which represents the last $n - p$ columns and we extend some basis vectors to span $\mathcal{X}$ which gives,

$$T^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

³This can be verified on MATLAB

Step 2. We compute the following matrices and partition them:

$$TAT^{-1} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1.0000 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1.0000 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1.0000 \\ 0 & 493.7629 & -3.5865 & -74.8161 & -1.7518 & 1.7906 \\ 0 & 597.3631 & -23.7073 & -82.8864 & -2.1875 & 2.4444 \\ 0 & -610.6139 & 83.3820 & 84.7249 & 2.4444 & -3.3478 \end{bmatrix}$$

$$TB = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 136.2256 \\ 150.9200 \\ -154.2677 \end{bmatrix} \quad CT^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Step 3. We pick L_2 such that $A_{22} + L_2A_{12}$ is stable. Using the `place()` function on MATLAB, we get the following Observer gains with the desired poles

$$\sigma_{observer} = \{-73, -74, -75\}$$

$$L_2 = \begin{bmatrix} 1.8161 & 1.7518 & -1.7906 \\ 82.886 & -71.812 & -2.4444 \\ -84.725 & -2.4444 & -71.652 \end{bmatrix}$$

Step 4. The minimal order observer is

$$\dot{z}(t) = Mz(t) + Nu(t) + Py(t)$$

$$\hat{x}(t) = Qz(t) + Ru(t) + Sy(t)$$

where

$$M := A_{22} + L_2A_{12}, \quad N := B_2 + L_2B_1, \quad P := (A_{21} + L_2A_{11} - A_{22}L_2 - L_2A_{12}L_2)C_1^{-1},$$

$$Q := T^{-1} \begin{bmatrix} 0 \\ I_{n-p} \end{bmatrix}, \quad R := 0, \quad S := T^{-1} \begin{bmatrix} C_1^{-1} \\ -L_2C_1^{-1} \end{bmatrix}.$$

giving

$$M = \begin{bmatrix} -73 & 0 & 0 \\ 0 & -74 & 0 \\ 0 & 0 & -75 \end{bmatrix} \quad N = \begin{bmatrix} 136.23 \\ 150.92 \\ -154.27 \end{bmatrix} \quad P = \begin{bmatrix} 132.57 & 621.64 & -134.3 \\ 6133.6 & -4716.8 & -204.59 \\ -6354.4 & -793.94 & -5290.5 \end{bmatrix}$$

$$Q = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad S = \begin{bmatrix} 1.0000 & 0 & 0 \\ 0 & 1.0000 & 0 \\ 0 & 0 & 1.0000 \\ -54.8161 & -1.7518 & 1.7906 \\ -82.8864 & 22.8125 & 2.4444 \\ 84.7249 & 2.4444 & 26.6522 \end{bmatrix}$$

For faster convergence of the estimates, the initial condition $z(0)$ is set in such a way that it accounts for zero initial velocity at the three joints at the near upright position (Quanser engages the control once α & ϕ are within 1.5°). From [1, p. 124], we have that:

$$z(t) := \hat{x}_2(t) + L_2 C_1^{-1} y(t)$$

Since $\hat{x}_2(t)$ is 0 at the near upright position, once the control engages, we reset the integral initial condition to

$$z(0) = L_2 C_1^{-1} y(0) \quad (10)$$

Where $y(0)$ is the position measurement at the time the control is engaged. Using the state-feedback gains from 8, we obtain the following simulated response for an arbitrary initial condition of:

$$x_o = [5 \quad -2 \quad 1 \quad 0 \quad 0 \quad 0]^T$$

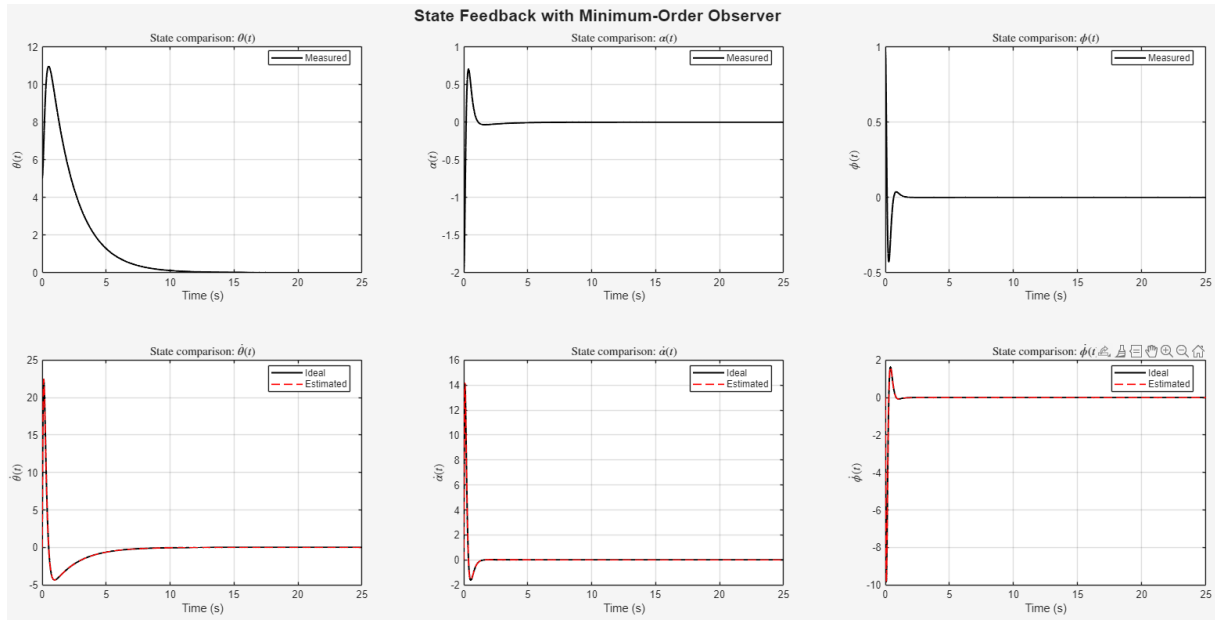


Figure 2: Minimum Order Observer with State Feedback Controller

8.3 Hardware Implementation

An unit delay block was introduced to the input of the observer because the build of the file created an algebraic loop.

9 Regulation and Tracking

This section deals with the tracking of a step reference signal and the regulation of the plant. The reference signals are assumed to satisfy known differential equations.

9.1 The Case of Full Information

Assuming full state feedback, we consider the case of full information⁴ for tracking. The setup is a plant with state x_1 and an exosystem with state x_2 as follows

$$\begin{aligned}\dot{x}_1 &= A_1x_1 + A_3x_2 + B_1u \\ \dot{x}_2 &= A_2x_2 \\ e &= D_1x_1 + D_2x_2 \\ y &= \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\end{aligned}$$

Combining the two states

$$\dot{x} = Ax + Bu, \quad e = Dx, \quad y = x,$$

$$A = \begin{bmatrix} A_1 & A_3 \\ 0 & A_2 \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ 0 \end{bmatrix}, \quad D = \begin{bmatrix} D_1 & D_2 \end{bmatrix}.$$

and with the controlled system ($u = Fx$)

$$\dot{x} = (A + BF)x, \quad e = Dx,$$

$$A + BF = \begin{bmatrix} A_1 & A_3 \\ 0 & A_2 \end{bmatrix} + \begin{bmatrix} B_1 \\ 0 \end{bmatrix} \begin{bmatrix} F_1 & F_2 \end{bmatrix} = \begin{bmatrix} A_1 + B_1F_1 & A_3 + B_1F_2 \\ 0 & A_2 \end{bmatrix}.$$

The regulation problem is one that solves the condition that $e(t)$ goes to zero for every $x(0)$. Thus we need to solve for the **Regulator equations** to see if it's solvable by a control of the form $u = Fx$

Here the state variables of the plant and the exomodel are:

$$x_1 := \begin{bmatrix} \theta \\ \alpha \\ \phi \\ \dot{\theta} \\ \dot{\alpha} \\ \dot{\phi} \end{bmatrix} \quad x_2 := \begin{bmatrix} r \end{bmatrix}$$

⁴This method doesn't work very well experimentally

The regulated variable is the tracking error

$$e := r - \theta$$

The controller has access to the full state of the pendulum and the reference signal

$$y := \begin{bmatrix} \theta \\ \alpha \\ \phi \\ \dot{\theta} \\ \dot{\alpha} \\ \dot{\phi} \\ r \end{bmatrix}$$

With the latter, we get the system of the form

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 493.7629 & -3.5865 & -74.8161 & -1.7518 & 1.7906 & 0 \\ 0 & 597.3631 & -23.7073 & -82.8864 & -2.1875 & 2.4444 & 0 \\ 0 & -610.6139 & 83.3820 & 84.7249 & 2.4444 & -3.3478 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 136.2256 \\ 150.9200 \\ -154.2677 \\ \hline 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Next we select F_1 such that to stabilize $A_1 + B_1 F_1$ which was already found from section 8

$$F_1 = \begin{bmatrix} -0.4870 & 24.6576 & 60.4800 & -0.6999 & 8.1836 & 7.7908 \end{bmatrix}$$

Next we solve for

$$A_1 X - X A_2 + A_3 + B_1 U = 0, \quad D_1 X + D_2 = 0$$

for X and U . Looking at the augmented matrices, we observe that

$$A_1 \in \mathbb{R}^{6 \times 6}, \quad A_2 \in \mathbb{R}^1, \quad A_3 \in \mathbb{R}^{6 \times 1}$$

Thus looking at the regulator equations, we deduce

$$X = \begin{bmatrix} x_{11} \\ x_{21} \\ x_{31} \\ x_{41} \\ x_{51} \\ x_{61} \end{bmatrix}, \quad U = [u_{11}]$$

After plugging it into the equation, MATLAB solves it for

$$X = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad U = [0]$$

The state feedback controller becomes

$$\begin{aligned} F &= [F_1 \quad F_2] = [F_1 \quad U - F_1 X] \\ &= [-0.48705 \quad 24.658 \quad 60.48 \quad -0.69993 \quad 8.1836 \quad 7.7908 \quad 0.48705] \end{aligned}$$

Where the first 6 gains are for the plant state variables and the last entry F_2 is for the step reference tracking. The exosystem is represented by

$$\dot{x}_2 = A_2 x_2$$

Where A_2 is zero, $B = 0$, $C = 1$, and $D = 0$, giving the constant output x_2 . The latter can be represented using a constant block and ensuring the value is converted into radians before being combined with x_1 .

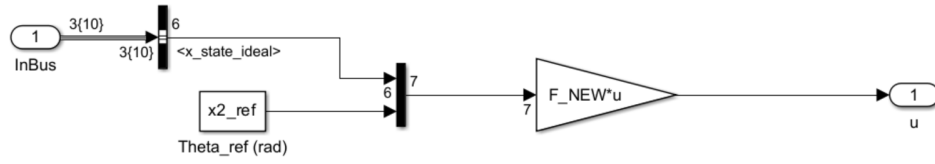


Figure 3: Reference Tracking Controller - Case of Full Information

With the latter controller, a demonstration of reference tracking is shown below

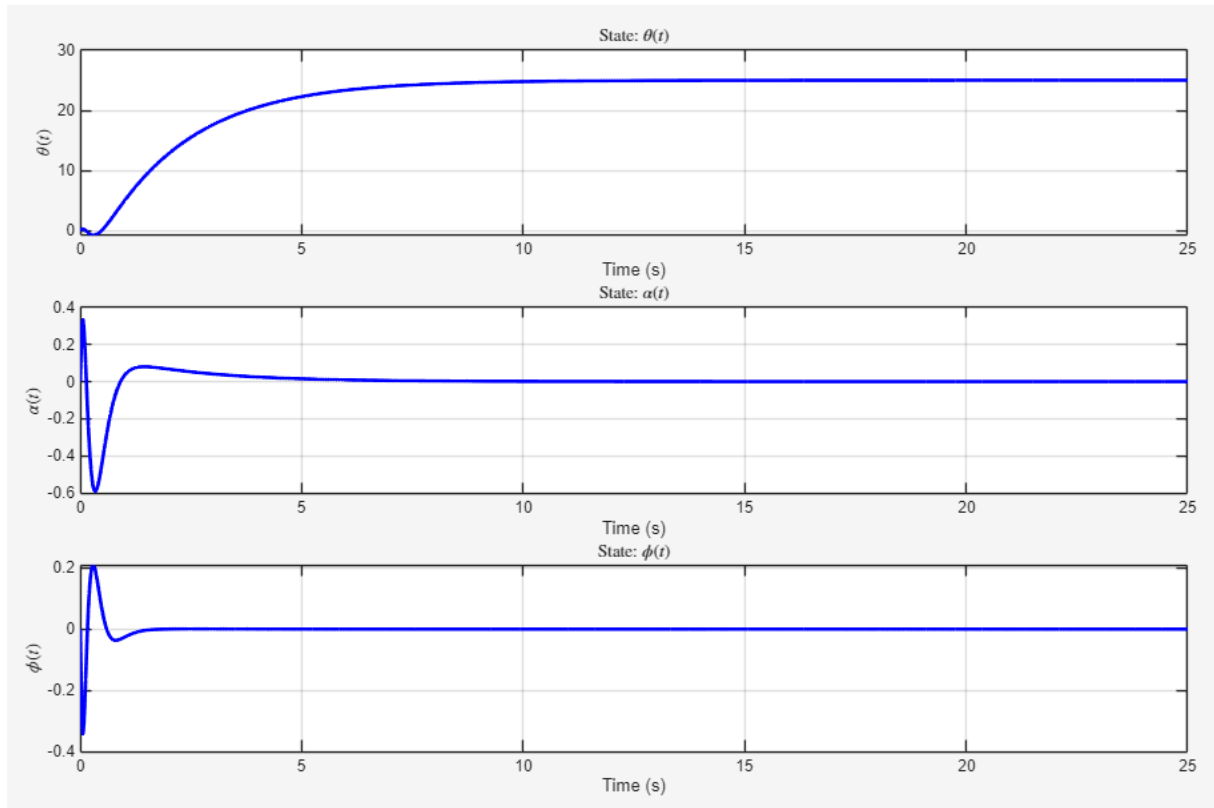


Figure 4: Reference Tracking Plot - Desired $\theta = 25$

9.2 The Case of Measurement Feedback

This section⁵ considers the case where the controller does not have sensors for the full state x_1 and x_2 of the exosystem. The augmented plant is similarly of the form.

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 493.7629 & -3.5865 & -74.8161 & -1.7518 & 1.7906 & 0 \\ 0 & 597.3631 & -23.7073 & -82.8864 & -2.1875 & 2.4444 & 0 \\ 0 & -610.6139 & 83.3820 & 84.7249 & 2.4444 & -3.3478 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 136.2256 \\ 150.9200 \\ -154.2677 \\ \hline 0 \end{bmatrix}$$

With

$$y = C_1 x_1 + C_2 x_2$$

$$C_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad C_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

and

$$e = D_1 x_1 + D_2 x_2$$

$$D_1 = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad D_2 = \begin{bmatrix} 1 \end{bmatrix}$$

We choose F to be the state feedback using the same methodology as in the previous subsection

$$F = \begin{bmatrix} -0.33398 & 18.351 & 51.673 & -0.332 & 6.647 & 6.5995 & 0.33398 \end{bmatrix}$$

We then choose L such that $A+LC$ is stable, and construct the observer-based controller with $\sigma(A+LC) = \{-161, -162, -163, -164, -165, -166, -167\}$

$$\dot{x}_c(t) = (A + BF + LC)x_c(t) - Ly(t)$$

$$u(t) = Fx_c(t).$$

⁵This method doesn't work very well experimentally

with

$$L = \begin{bmatrix} -253.78 & 1.62 & -1.1372 & 0 \\ 82.735 & -327.74 & -4.3533 & 7.1054 \times 10^{-15} \\ -83.64 & -3.7341 & -325.13 & -7.1054 \times 10^{-15} \\ -8300 & -69.142 & -378.61 & -4.5475 \times 10^{-13} \\ 20625 & -27235 & -981.73 & 0 \\ -20842 & -253.28 & -26075 & -1.819 \times 10^{-12} \\ -161 & 0 & 0 & -161 \end{bmatrix}$$

9.3 Structurally Stable Synthesis

This section deals with constructing a structurally stable synthesis. the controller has a state vector with two components $x_c = (x_{c1}, x_{c2})$ and the controller has the form

$$\begin{bmatrix} \dot{x}_{c1} \\ \dot{x}_{c2} \end{bmatrix} = \begin{bmatrix} A_{c1} & 0 \\ A_{c3} & A_{c2} \end{bmatrix} \begin{bmatrix} x_{c1} \\ x_{c2} \end{bmatrix} + \begin{bmatrix} B_{c1} \\ B_{c2} \end{bmatrix} y$$

$$u = \begin{bmatrix} C_{c1} & C_{c2} \end{bmatrix} \begin{bmatrix} x_{c1} \\ x_{c2} \end{bmatrix} + D_c y$$

with $y = (y_o, e)$. We have

$$C_1 = \begin{bmatrix} C_{01} \\ D_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad C_2 = \begin{bmatrix} C_{02} \\ D_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Thus giving

$$y = \begin{bmatrix} y_o \\ e \end{bmatrix} = \begin{bmatrix} \theta \\ \alpha \\ \phi \\ r - \theta \end{bmatrix}$$

Step 1: Denote the characteristic polynomial of A_2 (exomodel) by

$$\pi_2(s) = \det(sI - A_2) = \det(s - 0) = s$$

Where $n_2 = 1$, $\alpha_0 = 0$ and $q = 1$ (number of regulated variables. And so we set

$$A_{c1} = [0]$$

$$B_{ce1} = [1]$$

And we set

$$B_{c1} = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}$$

Step 2: Form the **augmented system** consisting of the plant and the portion of the controller from Step 1

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_{c1} \end{bmatrix} = \begin{bmatrix} A_1 & 0 \\ B_{c1}C_1 & A_{c1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_{c1} \end{bmatrix} + \begin{bmatrix} B_1 \\ 0 \end{bmatrix} u$$

$$y_a = \begin{bmatrix} C_1 & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} x_1 \\ x_{c1} \end{bmatrix}$$

And therefore

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_{c1} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1.0000 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1.0000 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1.0000 & 0 \\ 0 & 493.7629 & -3.5865 & -74.8161 & -1.7518 & 1.7906 & 0 \\ 0 & 597.3631 & -23.7073 & -82.8864 & -2.1875 & 2.4444 & 0 \\ 0 & -610.6139 & 83.3820 & 84.7249 & 2.4444 & -3.3478 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_{c1} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 136.2256 \\ 150.9200 \\ -154.2677 \\ 0 \end{bmatrix} u$$

$$y_a = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_{c1} \end{bmatrix}$$

We need to design a stabilizer for the augmented system choosing

$$F_a = \begin{bmatrix} F_{a1} & F_{a2} \end{bmatrix}$$

So that $A_a + B_a F_a$ is stable. We choose the poles $\sigma = \{-110, -9.85 + 1.62i, -9.85 - 1.62i, -6, -4.2, -0.5 + 0.27i, -0.5 - 0.27i\}$. We then choose L_1 so that $A_1 + L_1 C_1$ is stable with poles placed at $\sigma = \{-147, -149, -151, -153, -155, -157\}$ thus giving

$$L_1 = \begin{bmatrix} -114.57 & 1.55 & -1.61 & 114.57 \\ 44.67 & -298.51 & -4.64 & -44.67 \\ -40.78 & -4.11 & -304.01 & 40.78 \\ -3121.66 & -122.95 & -385.86 & 3121.66 \\ 9790.12 & -22680.28 & -908.40 & -9790.12 \\ -9217.63 & -224.96 & -22823.40 & 9217.63 \end{bmatrix}$$

Thus the controller has the form

$$\begin{bmatrix} \dot{x}_{c1} \\ \dot{x}_{c2} \end{bmatrix} = \begin{bmatrix} A_{c1} & 0 \\ A_{c3} & A_{c2} \end{bmatrix} \begin{bmatrix} x_{c1} \\ x_{c2} \end{bmatrix} + \begin{bmatrix} B_{c1} \\ B_{c2} \end{bmatrix} y$$

$$u = \begin{bmatrix} C_{c1} & C_{c2} \end{bmatrix} \begin{bmatrix} x_{c1} \\ x_{c2} \end{bmatrix} + D_c y$$

And replacing it with our numeric values we obtain

$$\begin{bmatrix} A_{c1} & 0 \\ A_{c3} & A_{c2} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -229.13 & 1.55 & -1.61 & 1.00 & 0 & 0 \\ 0 & 89.34 & -298.51 & -4.64 & 0.00 & 1.00 & 0 \\ 0 & -81.57 & -4.11 & -304.01 & 0 & 0 & 1.00 \\ 29.96 & -6354.94 & 3344.70 & 7158.40 & -156.87 & 1009.97 & 972.57 \\ 33.19 & 19456.56 & -18788.24 & 7429.91 & -173.79 & 1118.67 & 1077.94 \\ -33.92 & -18308.84 & -4203.32 & -31287.53 & 177.64 & -1143.27 & -1102.70 \end{bmatrix}$$

$$\begin{bmatrix} B_{c1} \\ B_{c2} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1.00 \\ 114.57 & -1.55 & 1.61 & -114.57 \\ -44.67 & 298.51 & 4.64 & 44.67 \\ 40.78 & 4.11 & 304.01 & -40.78 \\ 3121.66 & 122.95 & 385.86 & -3121.66 \\ -9790.12 & 22680.28 & 908.40 & 9790.12 \\ 9217.63 & 224.96 & 22823.40 & -9217.63 \end{bmatrix}$$

$$\begin{bmatrix} C_{c1} & C_{c2} \end{bmatrix} = \begin{bmatrix} 0.2199 & -0.8194 & 21.8306 & 55.4070 & -0.6023 & 7.4268 & 7.1263 \end{bmatrix}$$

$$D_c = \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$$

References

- [1] Dr. Christopher Nielsen. Multivariable control systems. *Waterloo University*, 2026. pages 13, 17
- [2] {Mark W.} Spong, Seth Hutchinson, and M. Vidyasagar. Robot modeling and control. *IEEE Control Systems*, 26(6):113–115, December 2006. pages 8